

CONJECTURE 1.7 OF QU AND ZANG IS FALSE

ABSTRACT. Let $b_{2,j}(n)$ be the total number of hooks of length j among the 2-regular partitions of n . We prove that, for every odd $n \geq 7$,

$$b_{2,n-3}(n) = 5, \quad b_{2,n-2}(n) = \frac{n-1}{2}.$$

Hence, for every even $k \geq 10$,

$$b_{2,k}(k+3) = 5 < \frac{k+2}{2} = b_{2,k+1}(k+3),$$

which disproves Conjecture 1.7 of Qu and Zang.

A t -regular partition is a partition with no part divisible by t . For $t \geq 2$ and $j \geq 1$, let $b_{t,j}(n)$ denote the total number of hooks of length j among all t -regular partitions of n . Singh and Barman conjectured that

$$b_{2,k}(n) \geq b_{2,k+1}(n) \quad (k \geq 3, n \geq 0, n \neq k+1)$$

[2, Conjecture 1.6]. Qu and Zang proved that this fails for every odd k , proved the cases $k = 4$ and $k = 6$, and conjectured the inequality for every even $k \geq 8$ [1, Theorems 1.4–1.6 and Conjecture 1.7]. The following elementary formulas disprove that conjecture for every even $k \geq 10$.

Theorem 1. *For every odd integer $n \geq 7$,*

$$b_{2,n-3}(n) = 5, \quad b_{2,n-2}(n) = \frac{n-1}{2}.$$

Consequently, for every even integer $k \geq 4$,

$$b_{2,k}(k+3) - b_{2,k+1}(k+3) = \frac{8-k}{2}.$$

In particular, the conjectured inequality fails for every even $k \geq 10$, and Conjecture 1.7 of Qu and Zang is false.

Proof. A 2-regular partition is a partition into odd parts. Let $\lambda \in \mathcal{P}_{\text{odd}}(n)$, written as $\lambda = (\lambda_1, \dots, \lambda_\ell)$. For a cell (r, c) of its Young diagram, define the defect by $d_\lambda(r, c) = n - h_\lambda(r, c)$. Counting the cells outside the hook of (r, c) gives

$$(1) \quad d_\lambda(r, c) = \sum_{i < r} \lambda_i + (c-1) + \sum_{i > r} (\lambda_i - \mathbf{1}_{\{\lambda_i \geq c\}}).$$

We claim that the complete lists of cells of defects 2 and 3 are as follows; repeated parts are written in exponential notation.

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Defect	Partition λ	Cell (r, c)
2	(n)	$(1, 3)$
	(1^n)	$(3, 1)$
	$(j, 3, 1^{n-j-3}), j \in \{3, 5, \dots, n-4\}$	$(1, 1)$
3	(n)	$(1, 4)$
	$(n-2, 1^2)$	$(1, 2)$
	$(3, 3, 1^{n-6})$	$(2, 1)$
	$(3, 1^{n-3})$	$(2, 1)$
	(1^n)	$(4, 1)$

We verify exhaustiveness. First suppose $r = 1$. Equation (1) implies $c \leq d_\lambda(1, c) + 1$. For defect 2, the cases $c = 3, 2, 1$ give, respectively, no lower row; exactly one lower part equal to 1, which would make $\lambda_1 = n - 1$ even; or

$$\sum_{i>1} (\lambda_i - 1) = 2.$$

The last equation forces exactly one lower part equal to 3, with all remaining lower parts equal to 1. This gives $(j, 3, 1^{n-j-3})$. Here j is odd and $n - j - 3$ is a nonnegative odd integer, so $j \in \{3, 5, \dots, n - 4\}$.

For defect 3, the cases $c = 4, 3, 2, 1$ give, respectively, no lower row; exactly one lower part equal to 1, again making λ_1 even; either two lower parts equal to 1 or one lower part equal to 3, of which only the former makes λ_1 odd; or an impossibility, since $\sum_{i>1} (\lambda_i - 1)$ is even. These are exactly the listed cells with $r = 1$.

Now suppose $r \geq 2$. If $c \geq 2$, then $\lambda_1 \geq \lambda_r \geq 3$, so

$$\sum_{i<r} \lambda_i + (c - 1) \geq \lambda_1 + 1 \geq 4.$$

Thus a cell of defect at most 3 must have $c = 1$. Set

$$U = \sum_{i<r} \lambda_i, \quad L = \sum_{i>r} (\lambda_i - 1).$$

Then $d_\lambda(r, 1) = U + L$, where L is even and

$$U \geq r - 1, \quad U \equiv r - 1 \pmod{2}.$$

If the defect is 2, these conditions force $r = 3$, $U = 2$, and $L = 0$; hence $\lambda = (1^n)$ and $(r, c) = (3, 1)$.

If the defect is 3, then U is odd, so r is even, and $r - 1 \leq U \leq 3$ gives $r \in \{2, 4\}$. For $r = 2$, if $\lambda_1 = 1$, then monotonicity gives $L = 0$ and the defect is 1. Thus $\lambda_1 = 3$, which forces $L = 0$ and $\lambda_2 \in \{1, 3\}$, giving $(3, 1^{n-3})$ and $(3, 3, 1^{n-6})$. For $r = 4$, one has $U = 3$ and $L = 0$, so $\lambda = (1^n)$. This proves the classification.

There are five partition–cell pairs of defect 3. For defect 2, there are two isolated pairs and $(n - 5)/2$ admissible values of j . Therefore

$$b_{2,n-3}(n) = 5, \quad b_{2,n-2}(n) = 2 + \frac{n-5}{2} = \frac{n-1}{2}.$$

Substituting $n = k + 3$ proves the remaining assertions. $\square$

Remark 2. Taking $k = 10$ gives the explicit counterexample

$$b_{2,10}(13) = 5 < 6 = b_{2,11}(13).$$

In the arXiv version of [1] (arXiv:2501.13753v1, p. 3), Conjecture 1.7 is introduced by the sentence “After checking $8 \leq k \leq 20$ and $n \leq 10000$, we propose the following conjecture...”; the pair $(k, n) = (10, 13)$ lies in the range so described. At $k = 8$, Theorem 1 gives equality at $n = 11$. Together with the results of Qu and Zang, $k = 8$ is the only fixed parameter $k \geq 3$ not decided by these results.

REFERENCES

- [1] W. Qu and W. J. T. Zang, *On the hook length biases of the 2- and 3-regular partitions*, J. Number Theory **280** (2026), 455–480. doi:10.1016/j.jnt.2025.08.016; arXiv:2501.13753.
- [2] G. Singh and R. Barman, *Hook length biases in ordinary and t -regular partitions*, J. Number Theory **264** (2024), 41–58. doi:10.1016/j.jnt.2024.05.001.